REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ONE WORLD, BISHOP ALPHA AND KINDOROKO SECONDARY SCHOOLS
FIRST FORM FOUR JOINT PRE – MOCK ASSESSMENT

033/1 **BIOLOGY 1**

(For Both Private and School Candidates)

Time: 2:30 Hours **Year: 2025**

MARKING GUIDE

1. **10 Marks @ 1**

SN	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
ANS	**E**	**C**	**B**	**D**	**A**	**B**	**C**	**A**	**B**	**C**

2. **06 Marks @ 1**

LIST A	(i)	(ii)	(iii)	(iv)	(v)	(vi)
LIST B	**C**	**G**	**E**	**A**	**F**	**B**

3. (a) ***Five importance of studying biology***: **(05 Marks @ 1)**

(i) *It helps us to understand ourselves better*. In biology, we learn about anatomy and morphology of our body and how different body organs and tissues functions.

(ii) *It helps us to become different professional people*. For example: some biologists become doctors, pharmacists, teachers and many other professions.

(iii) *It helps us to understand and conserve our environment*. For example; in biology, we learn about ecology on the relationship between living things and their environment. Therefore, by understanding the importance of the environment, people are able to conserve it.

(iv) *It helps us to know the causes, transmissions, signs and symptoms and prevention of diseases*. This helps us to be safe from diseases and improve our health.

(v) *It helps us to answer many fundamental questions*. There are many questions about the nature that are answered through studying biology. Example: Where did life come from? Is there any importance of death? Can people with black skin give birth to an albino child? Why can't we taste well food when suffering from flu (cold)? Can animals survive if there are no plants at all?

(vi) *It helps us to appreciate the nature*. Biology gives us the knowledge about many interesting things about different organisms. Example: Did you know that: - Its physically impossible for pigs to look up into the sky; Leeches have 32 brains and 2 hearts; Jellyfish do not have brain, heart or lungs; Slugs have 4 noses; Cats can taste the air; Horses cannot breathe through their mouth; Koalas can sleep up to 22 hours per day; Dolphins sleeps with one eye open; Giraffes sleep while standing up???????

(vii) *It helps us improve agricultural production*. Scientists use biology knowledge to produce hybrid plants and animals that are resistant to diseases and produce more yield.

(viii) *It encourages international cooperation*. This is because scientists must cooperate to conduct researches about different diseases like Covid-19, Ebola and HIV/AIDS.

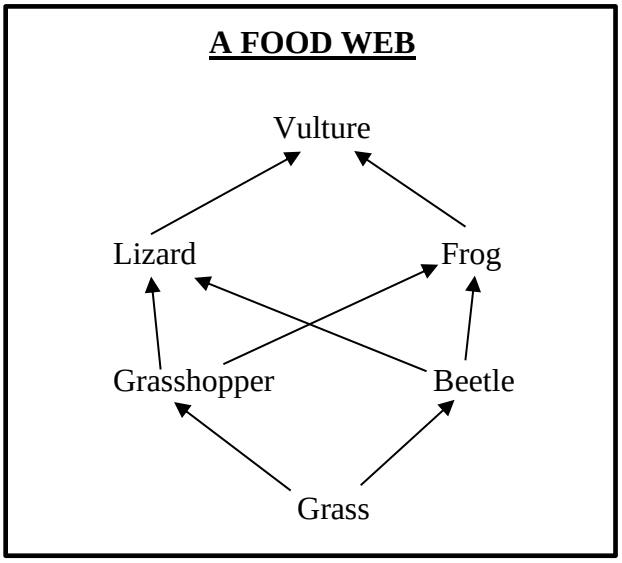
(ix) *It helps us to acquire research skills for carrying out scientific investigations*. These skills include measuring, observing, analyzing and making conclusion.

(x) *It is full of fun*. Biology is full of interesting activities. For example; most of biological experiments display colours which are beautiful and interesting.

(b) *Four differences between **mango cell** and **rod-shaped bacteria**.* **(04 Marks @ 1)**

MANGO CELL	ROD-SHAPED BACTERIA
It has true nucleus	It lacks true nucleus
It has membrane-bound organelles	It lacks membrane-bound organelles
It is larger in size	It is smaller in size
It has many cell organelles	It has few cell organelles
It has chloroplasts	It lacks chloroplasts (But may have chlorophyll in the cell membrane e.g., cyanobacteria)
It has linear DNA	It has circular DNA
It is non-motile (Lacks flagella or pilli)	Some are motile. (Have flagella or pilli)
Its chromosomes are wrapped by histones	Its chromosomes are not wrapped by histones
Its cell wall is made up of cellulose	Its cell wall is made up of peptidoglycan (murein)
It divides by mitosis or meiosis	It divides by binary fission

4. (a) **03 Marks**



(b) (i) Grass **(03 Marks @ 1)**

(ii) Lizard and Frog

(iii) Grasshopper and Beetle

(c) (i) Decomposers **(01 Mark)**

(ii) - Kingdom Fungi **(02 Marks @ 1)**

- Kingdom Monera

5. (a) (i) **Mitosis:** This is the process by which the cell divides to produce two identical daughter cells, each with the same number of chromosomes as the parent cell. **(02 Marks @ 1)**

(ii) *Four significances of mitosis*. **(04 Marks @ 1)**

- *It is a means of reproduction in organisms that reproduce asexually*. Unicellular organisms such as protozoans reproduce asexually through binary fission. Binary fission is through mitosis.
- *It is the basis of growth in multicellular organisms.* This is because it leads to the increase in number of cells; therefore, increase in the body size of an organism.
- *It is the means of cell replacement.* Damaged cells are usually replaced with new cells. This is possible because mitosis produces new cells that are exact copies of the cells being replaced.
- *It is a means of regeneration of body parts in some organisms.* Some animals can regenerate lost parts of their bodies through mitosis. Examples of such animals include lizards and some crustaceans.
- *It helps to retain the number of chromosomes hence support genetic stability*. Mitosis results in the production of two diploid daughter cells with similar genetic properties as their parent cell.

(b) *Three roles of placenta in humans soon after fertilization.* **(03 Marks @ 1)**

(i) It facilitates gaseous exchange to the fetus. The placenta is the only source of oxygen for the fetus.
(ii) It facilitates the transfer of food nutrients to the foetus. All nutrient for the fetus comes from mother through placenta.
(iii) It facilitates the excretion of waste products. It filters waste products from the fetus such as nitrogenous wastes.
(iv) It secretes of hormones. Example: Human Chorionic Gonadotrophin, Oestrogen and Progesterone.
(v) It aids the transfer of antibodies to the foetus. These provides the natural passive immunity
(vi) It prevents mixing of the foetal and maternal blood. This is because it has pores that allow the exchange of materials without actual blood mixing.

6. (a) Because a grasshopper obtains gases from the environment through spiracles and exchange gases through tracheal system. These structures are located on its abdomen and not on the head. **(01 Mark)**

(b) **Endotherms**: These are organisms whose body temperature remains constant despite of the changes in the environment temperature. Example: Birds and mammals like rats, humans, etc.

WHEREAS

Ectotherms: These are organisms whose internal body temperature changes with that of the environment. Example: fish, amphibians (like frogs) and reptiles (like lizards). **(02 Marks @ 1)**

(c) *Four ways through which water is lost from the human body*. **(06 Marks @ 1.5)**

(i) Urination. The kidneys filter out wastes and excess water, which is released as urine.
(ii) Sweating. The body releases sweat to cool itself, losing water in the process.
(iii) Breathing (Evaporation). Water vapor is lost when we exhale and from buccal and nasal cavity through evaporation.
(iv) Defecation. Some water is lost in stool, especially during diarrhea.
(v) Vomiting. This removes large amount of water from the body leading to dehydration.
(vi) Bleeding. Injuries or menstruation can cause water loss through blood.

Tears and salivation (A minor point). A small amount of water is lost through crying and saliva production.

7. (a) *Four consequences of not conducting proper cleanliness of the environment*. **(04 Marks @ 1)**

(i) *Eruption of diseases*. Poor environment may cause spread of diseases such as amoebic dysentery, cholera, typhoid and respiratory diseases.

(ii) *Environmental pollution*. Decaying waste may give gases that pollutes the air and cause unpleasant smell and respiratory diseases. Wastes also pollute the soil and degrade its fertility.

(iii) Loss of biodiversity. Accumulation of wastes leads to death of many organisms that are beneficial for proper recycling of nutrients in the ecosystem, especially organisms that lives in the soil and water bodies.

(iv) Destruction of the natural beauty of the environment. When wastes are not removed from the environment, they degrade the natural beauty of the environment. They make the environment unattractive.

(v) Occurrence of accidents. Accumulation of wastes in the environment may lead to unnecessarily accidents. People can be cut by broken glass or fall when they step on fruit peels such as bananas.

(vi) Spread of dangerous animals. When wastes accumulate for a long time, they create the habitat of dangerous animals such as snakes, bees and centipede. Most of these animals cause dangerous bites and stings which may result into severe pain or even death.

(vii) Blockage of the drainage channels. If wastes are not cleaned out, for example, used bottles and clothes left into water channels leads to the blockage of water channels. This can lead to flooding and hence severe effects to the people and other organisms.

(viii) Harm to the domestic animals. Plastic bags can choke or suffocate animals when they ingest them. If swallowed, the bags may block the digestive tract of the animal leading to death.

(ix) Breeding site for vectors. Wastes gives the site for breeding of the vectors such as mosquitoes, houseflies and cockroaches.

(b) (i) A nucleotide **(01 Mark)**

(ii) - DNA (Deoxyribonucleic acid) **(02 Marks @ 1)**
- RNA (Ribonucleic acid)

(iii) *Two differences between **DNA** and **RNA** basing on their parts Q and R* **(02 Marks @ 1)**

CRITERIA	DNA	RNA
Part **Q** (Pentose sugar)	It has deoxyribose sugar	It has ribose sugar **OR**
	It lacks oxygen atom at the second carbon of its pentose sugar	It has oxygen atom at the second carbon of its pentose sugar
Part **R** (Nitrogenous base)	It consists of thymine as one of its organic bases	It consists of uracil as one of its organic bases

8. (a) (i) Glucose/Sugar/Reducing sugar **(1.5 Marks)**

(ii) The patient is suffering from Glucosuria/Hyperglycemia/Diabetes mellitus **(1.5 Marks)**

(iii) Kidneys/Liver/Pancreas **(1.5 Marks)**

(b) *Three importance of movement to the living things*. **(4.5 Marks @ 1.5)**

(i) It helps them to escape from dangers, Example: fire, floods and predators.

(ii) It helps them to move to a better climatic condition; Example: birds migrate during extreme cold weather or drought while buffalo migrate to an area where they can get food and water during the dry season.

(iii) It helps them to search for food; Example: a lion chases an antelope, catches it and eat it.

(iv) It helps them to find mates so that they can reproduce.

(v) It helps them to avoid competition for living space and other life requirements

(vi) It helps plants to get maximum sunlight for photosynthesis

GUIDE FOR ESSAY QUESTIONS

Introduction:	2 Marks
Main body:	12 Marks @2 i.e.: Point=1, Explanation=1. (Total: 6 points)
Conclusion:	1 Mark
Grand Total:	**15 Marks**

9. ***Introduction:*** ***(Meaning of the brain and its structure)***

Brain is a specialized organ responsible for thinking, memory storage and coordinating the functions of many body organs. It is a collection of neurons located at the anterior end of an animal that act as a control center for many body processes. It is enclosed by a brain case called cranium in the skull. It is protected by three tough membranes called meninges. The outer part of the brain consists of grey matters (cerebral cortex) with cell bodies of neurons. The inner part consists of white matter with axon fibers of the neurons. The following are the adaptations of the brain.

Main Body: ***(Six adaptive features of the brain to its functions)***

(i) It is covered by hard skull bones ***for*** protection against physical injury.
(ii) It has highly convoluted tissues ***to*** increase the surface area for impulse processing.
(iii) It is covered by meninges with cerebrospinal fluid ***for*** protection against physical injury and shock
(iv) It has many relay neurons ***for*** linking information between the sensory and motor neurons.
(v) It has numerous mitochondria ***to*** generate energy for proper functioning.
(vi) It has a dense network of blood capillaries ***for*** transportation of oxygen, nutrients and wastes
(vii) It is connected to the spinal cord ***to*** coordinate functions of all other body parts

Conclusion: ***Any relevant. (For example: Ways to care for the brain)***

Generally, since the brain is a very delicate organ, people are advised to take care of the activities they do so as to avoid falling or harm to the head which may endanger the brain. Also, they should visit the health center whenever they feel unusual conditions of the head to monitor the brain health.

GUIDE FOR ESSAY QUESTIONS

Introduction:	2 Marks
Main body:	12 Marks @2 i.e.: Point=1, Explanation=1. (Total: 6 points)
Conclusion:	1 Mark
Grand Total:	**15 Marks**

10. ***Introduction: (Meaning of health and the concept of body defense/immunity)***

Health is a state of being well physically, mentally and socially. It is a general condition of the body and mind in which a person is free from disease, injury, damage or pain. **Immunity** is the ability of the body to resist infection. It is the ability of the body to defend itself against infectious agents, foreign cells and cancer cells. The immune system is the collection of cells and organs that are responsible to provide the body immunity. The following are some of the ways in which the body defends itself against different microorganisms that cause diseases.

Main Body: (Six ways in which the body defends itself against pathogens)

(i) ***The skin produces sweat and sebum*** which act as antimicrobial agent to kill some bacteria. Intact skin is also a physical barrier against entry of pathogens. Sweat glands in the skin produce sweat, which contains antibacterial agents that kill bacteria. Sebaceous glands of the skin produce oily substance called sebum which contains antibacterial agents that kills bacteria on the skin.

(ii) ***Eyes produce tears*** by the tear glands which act as antimicrobial agent to kill some bacteria. Tears contain enzymes which kill microorganisms and protect the eyeball from infection. Also, eye lashes, eyelids and eyebrows prevent dust particles and other foreign bodies from entering the eyes.

(iii) ***Ears produce wax*** which traps dust particles and some bacteria. The lining of the ear canal produces waxy material which traps bacteria, viruses andsmall dust particles.

(iv) ***The mouth produces saliva*** which act as antimicrobial agent to kill some bacteria. Salivary glands in the mouth produce saliva, which is a mild antibacterial agent.

(v) ***The stomach secretes concentrated hydrochloric acid*** which kill bacteria taken in with food. Hydrochloric acid produced by the stomach walls also destroys and kills most bacteria taken in with food. Digestive juices produced in the small intestine digest and kill the bacteria in the intestine.

(vi) ***The blood contains white blood cells*** which attack and destroy the pathogens. These include lymphocytes and phagocytes. The lymphocytes produce antibodies that destroy pathogens. The phagocytes engulf microorganisms invading the body, hence defending the body against disease – causing organisms.

(vii) ***The nose has hairs (cilia) and mucus*** to trap some dust particles and pathogens. Mucous secreting cells in the respiratory tract produce mucus which traps microorganisms and small dust particles. The respiratory tract such as trachea have hairs called cilia and mucus which trap and fitter dust and microorganisms.

Conclusion: Any relevant. (For example: Ways to improve body immunity)

Therefore, in order to improve body immunity, people are advised to take balanced diet every day, engage in regular physical exercises, complete treatment, practice proper personal hygiene and good manners and attend the health centers for regular check-up of their health.

GUIDE FOR ESSAY QUESTIONS

Introduction: 2 Marks
Main body: 12 Marks @2 i.e.: Point=1, Explanation=1. (Total: 6 points)
Conclusion: 1 Mark
Grand Total: 15 Marks

11. ***Introduction: (Meaning of transpiration)***

Transpiration is the process by which water is absorbed by plant roots from the soil, moves through the plant, and is released as vapor through pores, mainly in the leaves.

Main Body: (Three types of transpiration and Three significances of transpiration)

The following are three main types of transpiration:

(i) ***Stomatal Transpiration***: This is the most common type and occurs through small openings on the surface of leaves called stomata. The stomata are controlled by guard cells that regulate their opening and closing. Water vapor exits the plant when the stomata are open, usually during the day when photosynthesis occurs.

(ii) ***Cuticular Transpiration***: This type of transpiration takes place through the cuticle, a waxy layer covering the plant's surface, especially the leaves. It occurs when the stomata are closed, typically at night or during hot conditions. While less significant than stomatal transpiration, it still contributes to water loss.

(iii) ***Lenticular Transpiration***: This type occurs through the lenticels, which are small pores found in the bark of woody plants or stems. Lenticular transpiration is less common than stomatal and cuticular transpiration but still plays a role in the overall water loss from the plant.

The following are the significances of transpiration:

(i) ***Cooling the Plant***: Transpiration helps cool the plant by releasing water vapor, preventing overheating, especially in hot weather.

(ii) ***Transportation of Water and Nutrients***: Transpiration helps move water from the roots to the leaves, which is essential for the plant's growth and nutrient uptake. It also aids in transporting minerals and nutrients absorbed from the soil to different parts of the plant.

(iii) ***Maintaining Turgor Pressure***: Transpiration helps maintain the internal pressure in plant cells, keeping them firm and supporting the plant structure.

(iv) ***Regulating Water Balance***: Transpiration removes excess water from the plant, preventing waterlogging and root damage. This helps to balance water in the plant, ensuring it doesn't dry out or take in too much water.

(v) ***Maintaining Plant Shape***: Proper transpiration helps plants maintain their shape and structure, keeping them upright.

(vi) ***Preventing Pests and Disease***: The continuous movement of water through the plant can help remove waste products and prevent the buildup of harmful substances that may attract pests.

Conclusion: Any relevant. (For example: Ways to control the rate of transpiration)

Generally, the knowledge of biology on the concept of transpiration is very important, for example, farmers can increase yield in their farm by ensuring that plants in the farm has the balanced level of water. This can be achieved through controlling the rate of transpiration by watering the plants properly especially in the early and late hours of the day, shading the plants, mulching, pruning excess leaves, controlling the wind direction and speed, using proper plant spacing and improving soil quality.

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